



COVENANT UNIVERSITY

CANAANLAND, KM 10, IDIROKO ROAD
P.M.B 1023, OTA, OGUN STATE, NIGERIA.

TITLE OF EXAMINATION: B.SC EXAMINATION

COLLEGE: SCIENCE AND TECHNOLOGY

DEPARTMENT: COMPUTER AND INFORMATION SCIENCES

SESSION: 2024/2025

COURSE CODE: CSC317

COURSE TITLE: SYSTEM ANALYSIS AND DESIGN

INSTRUCTION: ANSWER ANY **THREE (3)** QUESTIONS

SEMESTER: ALPHA

CREDIT UNIT: 2

TIME: 2 HOURS

Question 1

- With the appropriate use of diagrams, list and explain the main cardinal relationships that can exist between entities (Using Engineering Information Style notation). (3 marks)
- How would you adequately explain the processes of creating a Process 0 diagram? Explain exhaustively with proper diagrams where appropriate. (3 marks)
- Provide two different options that are key for selecting an identifier for a student entity in Covenant University. Justify your choice for these options. (2 marks)
- A patient visits a hospital to use the patient billing system in order to make enquiries about the types of drugs he can buy. He must have insurance or he won't be attended to by the Nurse. The nurse checks his vitals and refers him to a doctor who in turn refers him to the Insurance Carrier to confirm the patient has his insurance covered. On confirmation, the doctor advises the patient on which drugs to use for his condition.
Note that a patient must be assigned to only one doctor but a doctor can have many patients. A patient can have only one insurance ID to prevent insurance fraud.
 - Draw an ERD to model the relationships between the entities identified. (2 marks)
 - Data Flow Diagram (DFD):
 - Draw a Context diagram to model the entire system. (3 marks)
 - Draw a Level 0 DFD to show various processes involved. (3.5 marks)
 - Draw a Level 1 DFD to model the process of drug prescription. (3 marks)
- While explaining to your boss about hosting his newly acquired information system, you mentioned to him about using the Client-Server Tier. Explain in details the various Client-Server Tiers you know in order to properly convince him. (3 marks)
- 'A Data Flow diagram shows data movement from one system entity to another system entity'. Explain this sentence with an appropriate example. (1 mark)



Question 2

- What are some mechanical errors that can result while constructing a Dataflow diagram? Outline and explain them with the aid of appropriate diagrams. (3 marks)
- Define what is meant by a *Relationship* in a data model. How should a relationship be named? (1 mark)
- After constructing the DFD and tackling the mechanical errors, what are those ways you can ensure the validity and integrity of the diagram? (3 marks)
- A cinema wants to manage its movie ticketing system efficiently. The system should keep track of movies, ticket details, and customers who purchase tickets. Movies are identified by their title and have a producer, length, director, and genre. Tickets are issued for a specific movie and have attributes such as price, whether it's for an adult or child and the showtime.
 - Draw an ERD to model the relationships between Movie, Ticket, and customer (3 marks)
 - Data Flow Diagram (DFD):
 - Draw a Context diagram to model the entire system. (2 marks)
 - Draw a Level 0 DFD to show various processes involved. (3.5 marks)
 - Draw a Level 1 DFD to model the process of purchasing a movie ticket. (3 marks)
- Your boss has asked you to develop an application that will function like Joomla and you do not have the functional or technical experience to handle this task. What system acquisition strategy do you choose instead? Enumerate any four other factors that supports your choice. (4 marks)
- 'An Entity Relationship diagram shows data movement from one entity to a process and to another entity.' Explain this sentence with an appropriate example. (1 mark)

Question 3

- With the aid of a suitable diagram, illustrate the relationship in a use case diagram. (5 marks)
- What kinds of Relationships exist between the following?
 - A Plane and a Pilot. (2 marks)
 - An Operating System and a Desktop Computer. (2 marks)
- Differentiate between an object and a class using suitable example. (2 marks)
- An online university registration system enables the staff of each academic department to examine the courses offered by their department, add and remove courses, and change the information about them (e.g., the maximum number of students permitted). It should permit students to check currently available courses, add and drop courses to and from their schedules, and examine the courses for which they are enrolled. Department staff should be able to print a variety of reports about the courses and the students enrolled in them. The system should ensure that no student takes too many courses and that students who have any unpaid fees are not permitted to register. (Assume that a fees data store is maintained by the university's financial office, which the registration system accesses but does not change).
 - Create a use case diagram for this scenario. (7 marks)
 - Draw a class including at least two (2) attribute and 2 methods of each class. (7.5 marks)

Question 4

- Differentiate between tools and techniques in system development and give three (3) examples of each. (5 marks)
- With the aid of diagrams, explain the following class relationships in system analysis:
 - (i) Polymorphism, (ii) Composition, (iii) Aggregation, and (iv) Association. (6 marks)
- An online university registration system enables the staff of each academic department to examine the courses offered by their department, add and remove courses, and change the information about them (e.g., the maximum number of students permitted). It should permit students to check currently available courses, add and drop courses to and from their schedules, and examine the courses for which they are enrolled. Department staff should be able to print a variety of reports about the courses and the students enrolled in them. The system should ensure that no student takes too many courses and that students who have any unpaid fees are not permitted to register. (Assume that a fees data store is maintained by the university's financial office, which the registration system accesses but does not change).
 - Using this scenario, draw two activity diagrams; each for student and academic staff interaction with the system. (7.5 marks)
 - Draw sequence diagram for student's interaction with the system. (5 marks)

Question 5

- You have been hired as the project manager for a company planning to launch an e-commerce platform to sell handmade crafts. During the planning phase of the project, the stake-holders express concerns about the project's feasibility and ask you to provide a detailed analysis.
- Outline the activities you would undertake in the project planning phase to address their concerns and prepare the project for execution. (5 marks)
 - Outline the activities that must be completed during the design phase and implementation phase. (5 marks)
 - Explain the four installation strategies available for system implementation, based on their characteristics. (5 marks)
 - Explain the terms Encapsulation, Inheritance and Multiple Inheritance, with respect to Object Orientation. (5 marks)
 - State three (3) drawbacks of multiple inheritance. (3.5 marks)